Arbitrary-prefix same-depth inverse rigidity on a record-atom tail

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For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For an exponent word $w = (a_1, \ldots, a_L)$, define

$$\mathrm{B}(w) := \sum_{i=1}^{L} a_i, \qquad \mathrm{C}(w) := \sum_{i=1}^{L} 3^{L-i} 2^{a_1+\cdots+a_{i-1}}. \tag{2}$$

Thus an exact path with word w satisfies

$$2^{\mathrm{B}(w)} U_+^L(x) = 3^L x + \mathrm{C}(w). \tag{3}$$

Use the record-atom notation

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{4}$$

The purpose of this note is to classify the least same-depth inverse over an arbitrary finite word in these atoms. No bound on the number of atoms or on the intervening runs of A is imposed.

1 Shifted carries

It is convenient to translate every state by one. Define

$$\mathrm{D}(w) := \mathrm{C}(w) + 2^{\mathrm{B}(w)} - 3^{|w|}. \tag{5}$$

If $X = x + 1$ and $Y = U_+^{|w|}(x) + 1$, then (3) becomes

$$2^{\mathrm{B}(w)} Y = 3^{|w|} X + \mathrm{D}(w). \tag{6}$$

Lemma 1 (nonnegative and uniformly bounded shifted carry)**.** *For every positive exponent word $v = (d_1, \dots, d_\ell)$,*

$$\mathrm{D}(v) = \sum_{i=1}^{\ell} 3^{\ell-i} 2^{d_1+\cdots+d_{i-1}} (2^{d_i} - 2) \geq 0 \tag{7}$$

and

$$\frac{\mathrm{C}(v)}{2^{\mathrm{B}(v)}} \leq \sum_{j=0}^{\ell-1} \frac{3^j}{2^{j+1}} = \left(\frac{3}{2}\right)^{\ell} - 1. \tag{8}$$

If W is a nonempty finite word over the atoms A, B, C, D, then

$$\boxed{0 \leq \mathrm{D}(W) < 4\,3^{|W|}.} \tag{9}$$

Proof. Expanding (5) one letter at a time gives (7). Also

$$\frac{\mathrm{C}(v)}{2^{\mathrm{B}(v)}} = \sum_{i=1}^{\ell} \frac{3^{\ell-i}}{2^{d_i+\cdots+d_\ell}},$$

and $d_i + \cdots + d_\ell \geq \ell - i + 1$, proving (8).

For concatenated words,

$$\mathrm{D}(UV) = 3^{|V|}\,\mathrm{D}(U) + 2^{\mathrm{B}(U)}\,\mathrm{D}(V). \tag{10}$$

The atom data are

P	$\lvert P\rvert$	$\mathrm{B}(P)$	$\mathrm{C}(P)$	$\mathrm{D}(P)$
A	1	1	1	0
B	2	3	5	4
C	4	6	65	48
D	4	6	73	56.

(11)

Write $h(W) = \mathrm{D}(W)/3^{|W|}$. From (10),

$$h(PU) = \frac{\mathrm{D}(P)}{3^{|P|}} + \frac{2^{\mathrm{B}(P)}}{3^{|P|}} h(U). \tag{12}$$

If $h(U) < 4$, the four right sides in (12) are respectively less than

$$\frac{8}{3}, \qquad \frac{4}{9} + \frac{32}{9} = 4, \qquad \frac{48 + 256}{81}, \qquad \frac{56 + 256}{81}.$$

They are all at most 4, with strictness inherited in the B case. Induction on the number of atoms proves (9). □

2 Distortion states

Let W be an exact record-atom word from x to y, and let V be an exact positive exponent word of the same raw length from z to y. At any common raw-time boundary put

$$\Delta := \mathrm{B}(V_{\mathrm{rem}}) - \mathrm{B}(W_{\mathrm{rem}}), \qquad z = 2^{\Delta}x + c, \qquad u := \frac{c}{2^{\Delta}}, \tag{13}$$

where "rem" denotes the two remaining suffixes from that boundary to the common endpoint. At the endpoint the state is $(\Delta, c) = (0, 0)$.

Suppose an actual atom P of length ℓ, mass b, and carry C_P is prepended to a processed suffix. Partition the alternative path at the same raw time and let v be the corresponding length-ℓ block, with mass s and carry C_v. If (Δ, c) is the state on the right and (Δ', c') the state on the left, then

$$\boxed{\Delta' = \Delta + s - b, \qquad c' = \frac{2^{\Delta'}C_P + 2^{s}c - C_v}{3^{\ell}}.} \tag{14}$$

Consequently, with $R(v) := C_v/2^s$,

$$\boxed{u' = \frac{2^b}{3^\ell}u + \frac{C_P}{3^\ell} - \frac{2^{b-\Delta}}{3^\ell}R(v).} \tag{15}$$

Because both boundary states are odd, every genuine state lies in the parity lattice

$$\begin{cases} c \in 2\mathbb{Z} + 1, & \Delta \geq 1, \\ c \in 2\mathbb{Z}, & \Delta = 0, \\ c = m/2^{-\Delta} \text{ with } m \text{ odd}, & \Delta < 0. \end{cases} \tag{16}$$

The only finite arithmetic needed below is isolated next. It is included as an exact certificate rather than hidden in a search over word length.

Lemma 2 (exceptional transition certificate)**.** *For a lattice-admissible left state with $\Delta' \geq -1$, the eight exceptional cases have the following sharp bounds; a dash means none exists.*

(Δ, c)	P	$\max R(v)$	*maximizing* v
$(-1, -\frac{1}{2})$	B	–	–
$(0,0)$	B	$5/8$	12
$(-1, -\frac{1}{2})$	C	$85/512$	1215
$(-1, -\frac{1}{2})$	D	–	–
$(0,0)$	C	$73/32$	1121
$(0,0)$	D	$73/64$	1122
$(1,-1)$	C	–	–
$(1,-1)$	D	$85/256$	1214.

(17)

If the right state satisfies $\Delta \geq -1$ and $-1 \leq u \leq 5$, with $c = -1/2$ when $\Delta = -1$, then no lattice-admissible transition has $\Delta' < -1$. Moreover, among all strip states which enter the half-state $(-1, -1/2)$ in one atom, the complete preimage table is

$(\Delta, c)_{\text{right}}$	P	v	$(\Delta', c')_{\text{left}}$
$(-1, -\frac{1}{2})$	A	1	$(-1, -\frac{1}{2})$
$(0,0)$	C	1121	$(-1, -\frac{1}{2})$.

(18)

Proof. For $\Delta' \leq 2$, the total mass $s = \Delta' - \Delta + b$ is fixed. Substitution of all positive compositions of s into (14) gives the low part of (17).

For $\Delta' \geq 3$, write $v = (a, \eta)$. In all eight cases the numerator in (14) is an integer of the form

$$2^a K(\eta) - 3^{\ell-1}. \tag{19}$$

Thus divisibility by 3^ℓ depends on a only through its parity, because increasing a by two changes (19) from N to $4N + 3^\ell$. The suffix exponents may be reduced to positive residues modulo

$$\operatorname{ord}_9(2) = 6, \qquad \operatorname{ord}_{81}(2) = 54. \tag{20}$$

Indeed, transfer of one full period from a suffix exponent to a preserves the total mass and the congruence, and can only increase

$$R(v) = \sum_{i=1}^{\ell} \frac{3^{\ell-i}}{2^{d_i + \cdots + d_\ell}}. \tag{21}$$

For each of the two parities, lowering a to the first positive value with $\Delta' \geq 3$ again preserves the congruence and increases (21). It therefore suffices to inspect 6 suffix residues for B and 54^3 suffix triples for C, D, with two choices of parity. Exact integer substitution gives (17).

The cases which could have $\Delta' < -1$ have total mass at most five; direct substitution in (14) gives no point of the lattice (16). Finally, imposing $(\Delta', c') = (-1, -1/2)$ bounds the right defect by 1 and the alternative mass by 6. Substitution gives exactly the two rows of (18). The accompanying program `verify_divergence_record_arbitrary_prefix.py` performs these finite integer checks and independently verifies every maximum in (17). □

Lemma 3 (uniform distortion strip and rigidity at defect -1)**.** *At every atom boundary of the two aligned paths,*

$$\boxed{\Delta \geq -1, \qquad -1 \leq u \leq 5.} \tag{22}$$

If the remaining actual suffix is nonempty and $\Delta = -1$, then

$$\boxed{c = -\frac{1}{2}, \qquad z = \frac{x-1}{2}.} \tag{23}$$

Proof. We induct from the common endpoint toward the two starts. The empty suffix has $(\Delta, c, u) = (0, 0, 0)$.

First consider the upper edge of the strip. Since $C_v > 0$, equation (15) and $u \leq 5$ give

$$u' < \frac{5\,2^b + C_P}{3^\ell} \leq 5; \tag{24}$$

the four numerators for A, B, C, D are $11, 45, 385, 393$, while the four denominators are $3, 9, 81, 81$.

For the lower edge, (8) gives $R(v) \leq 1/2, 5/4, 65/16$ for blocks of length $1, 2, 4$, respectively. Substitution in (15) proves $u' \geq -1$ directly for

$$A : \Delta \geq -1, \qquad B : \Delta \geq 1, \qquad C, D : \Delta \geq 2. \tag{25}$$

At the smallest listed defects the resulting lower bounds are

$$-1, \qquad -\frac{8}{9}, \qquad -\frac{64}{81}, \qquad -\frac{56}{81}.$$

It remains to treat lower right defects. By the inductive form of (23), $\Delta = -1$ has $c = -1/2$. If $\Delta = 0$, the lattice and the strip give $c \in \{0, 2, 4\}$; if $\Delta = 1$, they give $c \in \{-1, 1, 3, 5, 7, 9\}$. The worst values $c = -1/2, 0, -1$ are precisely the rows of (17). Equation (15) then gives, in the nonempty cases, the lower bounds

$$0, -\frac{1}{4}, -1, 0, \frac{3}{8} \tag{26}$$

for $B(0), C(-1), C(0), D(0), D(1, -1)$. For $c \geq 2$ at defect zero and $c \geq 1$ at defect one, the unrestricted bound (8) already gives a value greater than -1. The finite below--1 assertion of Lemma 2 proves $\Delta' \geq -1$. This establishes (22).

Suppose now that a nonempty processed suffix W has $\Delta = -1$. The lattice and (22) give

$$2c \in \{-1, 1, 3, 5\}. \tag{27}$$

Put $X = x + 1$, $Z = z + 1$, and

$$k := -c - \frac{1}{2}, \qquad Z = \frac{X}{2} - k. \tag{28}$$

Since the alternative suffix has one less unit of mass, its shifted equation (6), multiplied by two and compared with the actual equation, gives

$$\mathrm{D}(V) = \frac{\mathrm{D}(W)}{2} + k3^{|W|}. \tag{29}$$

By Lemma 1, $\mathrm{D}(V) \geq 0$ and $\mathrm{D}(W) < 4\,3^{|W|}$. Therefore $k \leq -2$ is impossible. The value $k = -1$ corresponds to $c = 1/2$, but every atom begins with exponent one, so $x \equiv 3 \pmod 4$; then $z = (x + 1)/2$ is even. The only remaining value is $k = 0$, which is (23). This also closes the inductive assertion used above. □

3 The arbitrary-prefix minimum

Theorem 4 (arbitrary-prefix same-depth inverse minimum)**.** *Let*

$$W = PU, \qquad P \in \{B, C, D\}, \qquad U \in \{A, B, C, D\}^*, \tag{30}$$

be an exact word from a positive odd integer q to y, and put $L = |W|$. Among all positive odd L-step predecessors of y, the unique least one is

$$\boxed{q_{\min} = \begin{cases} q, & P = B \text{ or } D, \\ (q-1)/2, & P = C. \end{cases}} \tag{31}$$

In the second case the unique minimum word is $1121\,U$*; it is exactly the local half-coalescing branch at the initial atom* C*, followed by the unchanged suffix* U*.*

Proof. Let V be any exact length-L word from z to y, and form its start state (Δ, c). Lemma 3 gives $\Delta \geq -1$.

If $\Delta = -1$, equation (23) gives $z = (q-1)/2$. Read the first actual atom forward, or equivalently use the one-atom preimage table (18). Since the left state is the half-state, the first atom is either A and the half-state persists, or it is C, the alternative block is 1121, and the right state is $(0, 0)$. Here the first atom belongs to $\{B, C, D\}$, so only the second alternative is possible. Thus $P = C$. At the right boundary the two positive odd states are equal. The accelerated map is deterministic, so the remaining exact word is U on both paths. Conversely, exactness of C gives $q \equiv 79 \pmod{128}$; direct substitution then gives the exact word 1121 from $(q-1)/2$ and the identity

$$U_+^4\left(\frac{q-1}{2}\right) = U_+^4(q) \tag{32}$$

for the words 1121 and 1113 shows that this branch always exists when $P = C$.

If $\Delta = 0$, the lattice and (22) give $c \in \{0, 2, 4\}$, hence $z = q + c \geq q$. Equality forces the same start and therefore the same deterministic word $V = W$. If $\Delta \geq 1$, (22) gives

$$z = 2^\Delta q + c \geq 2^\Delta (q-1) > q, \tag{33}$$

because an atom start has $q \equiv 3 \pmod 4$ and hence $q \geq 3$. These alternatives prove (31) and uniqueness. □

Corollary 5 (no finite same-depth pullback creates a new descent)**.** *Suppose an interior record tail contains a finite block beginning at a nontrivial atom* $P \in \{B, C, D\}$ *and ending at any later atom boundary. Extending the block across arbitrarily many intervening record atoms can never create a new same-depth smaller predecessor. If* $P = B$ *or* D*, the actual block start remains uniquely least. If* $P = C$*, the unique smaller predecessor is only the local half-seed at that initial* C*.*

Consequently, pulling a terminal-C *half-coalescence across any finite number of preceding record atoms cannot yield a new same-depth descent. Any genuinely new descent on this branch must change the raw inverse depth.*

Proof. Apply Theorem 4 to the complete finite atom word. Its two alternatives depend only on the first nontrivial atom, not on the length or composition of the suffix. □